

Cycle Lane Recommendation for Kingsway East

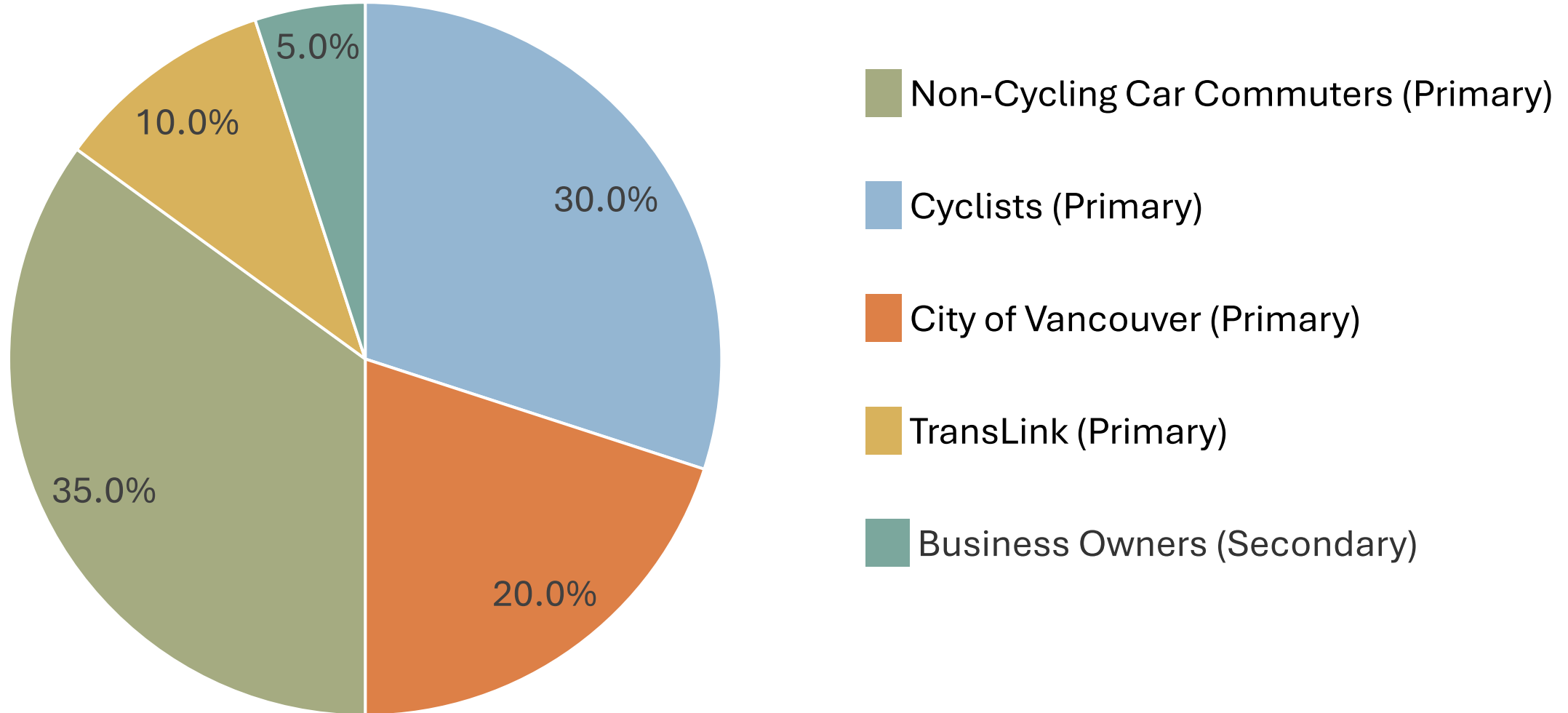
Agenda

- Overview and Recommendation
- Stakeholders
- Evaluation Criteria
- Decision Making
 - Screening
 - Most promising solution
 - Sensitivity analysis
- Project implementation
- Benefits and Disadvantages
- Conclusion

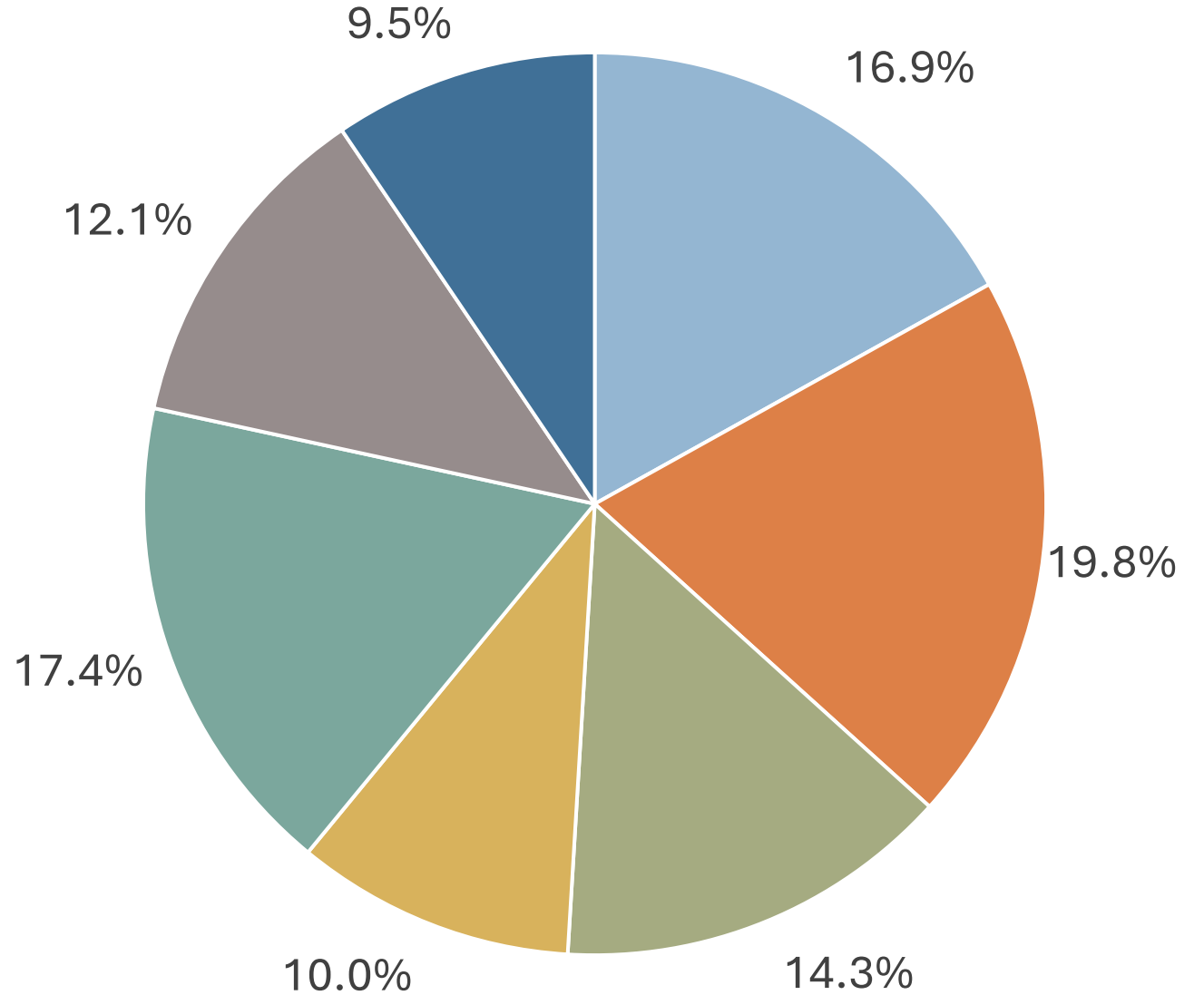
Overview and Recommendation

- Gas and diesel pollution
- Unsuccessful with increasing cycling
- **Goal:** Promote cycling
- **Recommendation:** Kingsway East

Importance Of Stakeholders



Evaluation Criteria



Screening

~~West Kent Ave
(Concrete Planters)~~

Greenway Project
(Concrete Planters)

~~West 14th Ave
(Concrete Planters)~~

Kingsway-West
(Concrete Planters)

Kingsway-East
(Concrete Planters)

~~West Kent Ave
(Concrete Barriers)~~

~~Greenway Project
(Concrete Barriers)~~

~~West 14th Ave
(Concrete Barriers)~~

~~Kingsway-West
(Concrete Barriers)~~

~~Kingsway-East
(Concrete Barriers)~~

~~West Kent Ave
(Bollards)~~

Greenway Project
(Bollards)

~~West 14th Ave
(Bollards)~~

Kingsway-West
(Bollards)

Kingsway-East
(Bollards)



The Most Promising Solution is Building a Bike Lane on Kingsway-East Using Concrete Barriers

A. Greenway Project
(Concrete Planters)



B. Kingsway-West
(Concrete Planters)



C. Kingsway-East
(Concrete Planters)



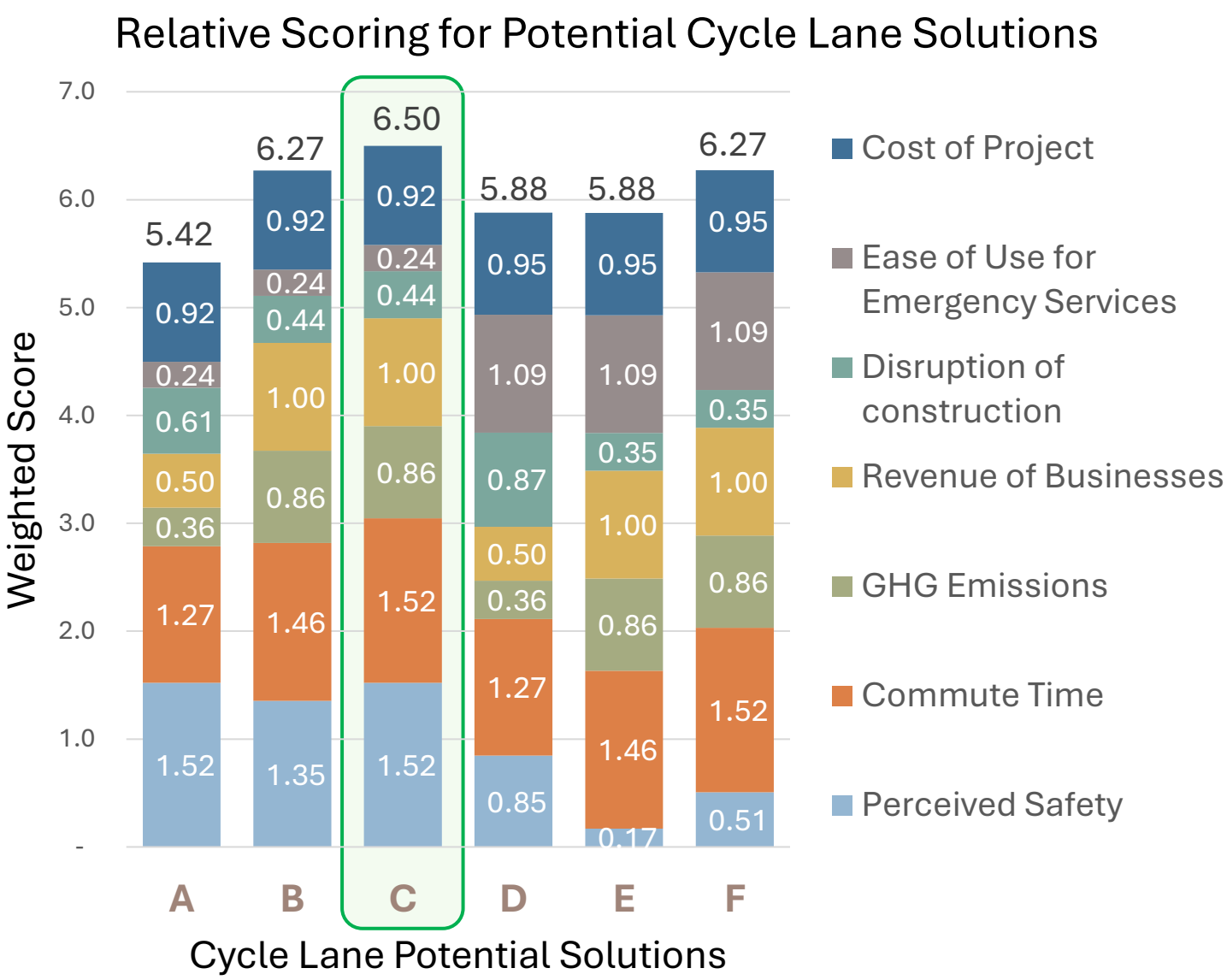
D. Greenway Project
(Bollards)



E. Kingsway-West
(Bollards)



F. Kingsway-East
(Bollards)



Sensitivity Analysis

- Solution C (Kingsway East Project-Concrete Planters) remained top (high robustness).
- "Perceived Safety"/"Commute Time" were most sensitive to scores.
- "GHG Emissions"/"Emergency Services ease" had little ranking impact.

Project Implementation – Kingsway East Concrete Planters

- Cost / km per Average Number of Cyclists Each Month: \$ 43.54
 - Cost / km: \$ 1.13M
 - Estimated Average Number of Cyclists per Month: 26 000
 - Distance: 6.8 km
 - Total Cost: \$ 7.7M
- Estimated GHG Reduction: 25%
 - Estimated Increase in Ridership: 246% (7500 – 26000 bikers)
 - GHG Combined Before Bike Lane: 1.045×10^9 g
 - After bike lane: 7.845×10^8 g (25% reduction of cars)

Assumptions

- Reduction of Car by 25%
- Roads with preexisting bike lane will receive less benefit
- Determine using CLD

Benefits

- Commute time
- Business revenue
- Perceived safety

Disadvantages

- Inhibited ease of use for emergency services
- Disruption of traffic by construction (busy road)
- More expensive than other options

Conclusion

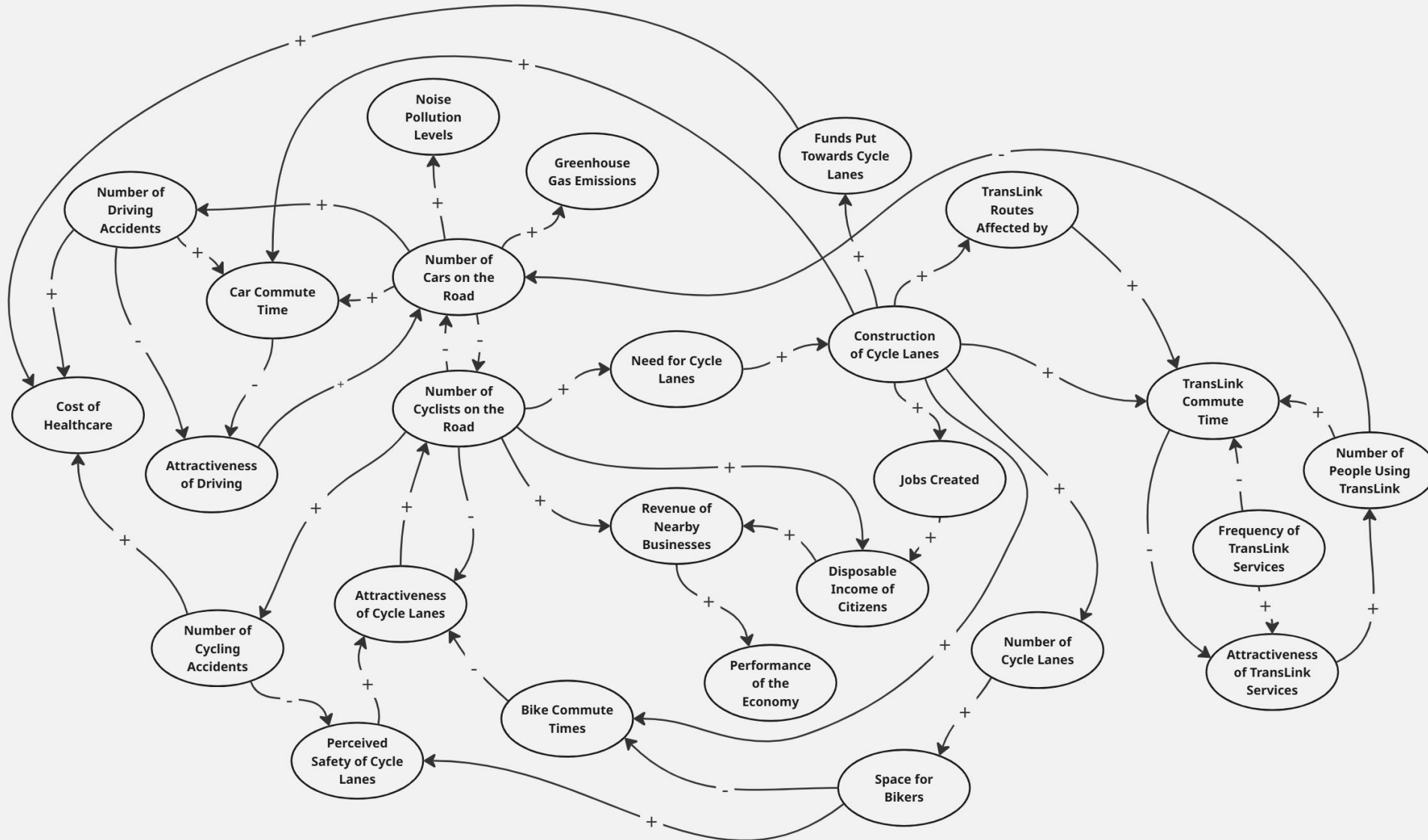
- Most suitable candidate: Kingsway East
- Scores the highest
- Safety, commute time

Thank You for Listening

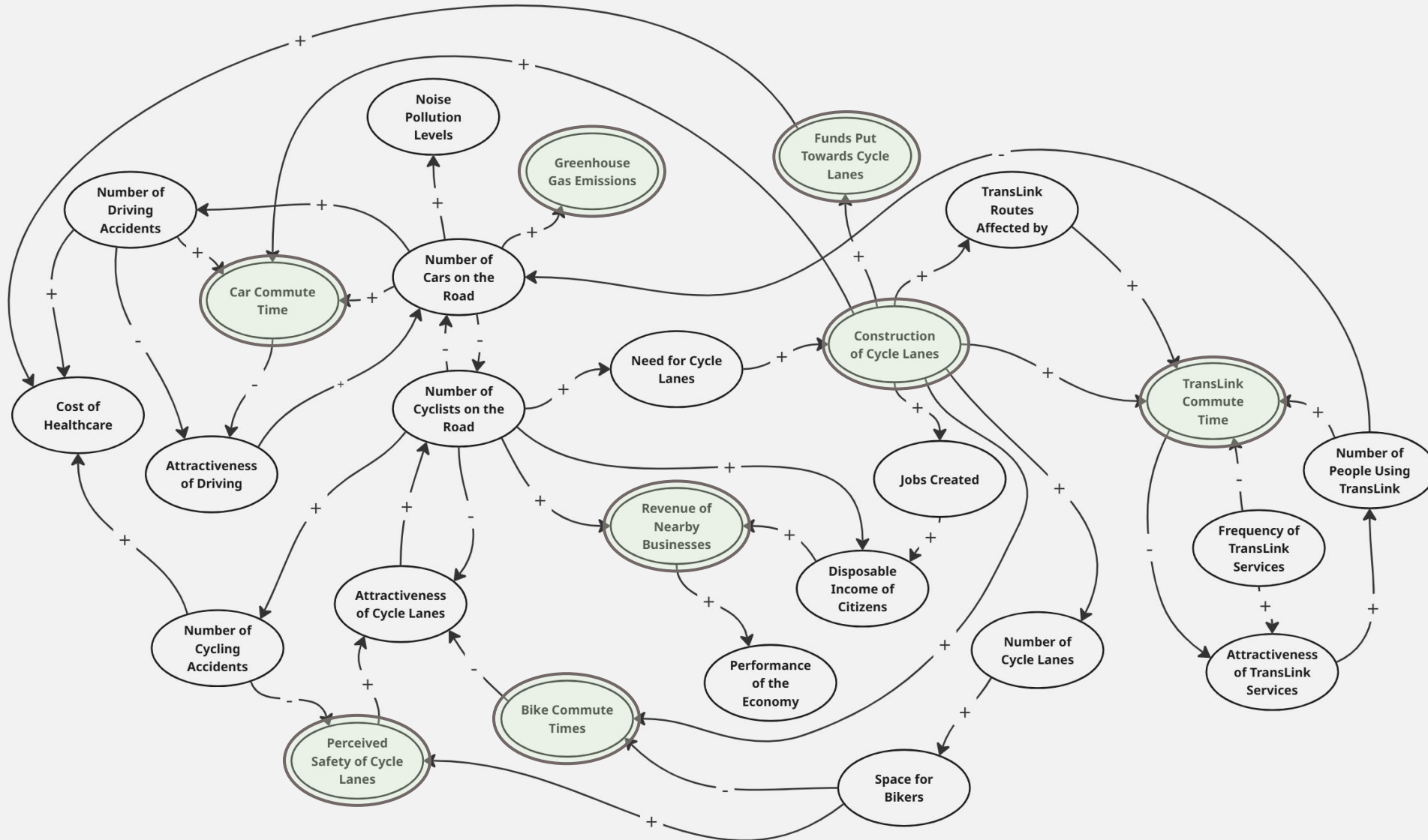
Q & A

Appendices

Appendix A – Causal Loop Diagram



Appendix A – Causal Loop Diagram



**Highlighted
Nodes**
Represent
Relations to Key
Specifications

Appendix B – Weighted Decision Matrix – Stakeholders (Part 1)

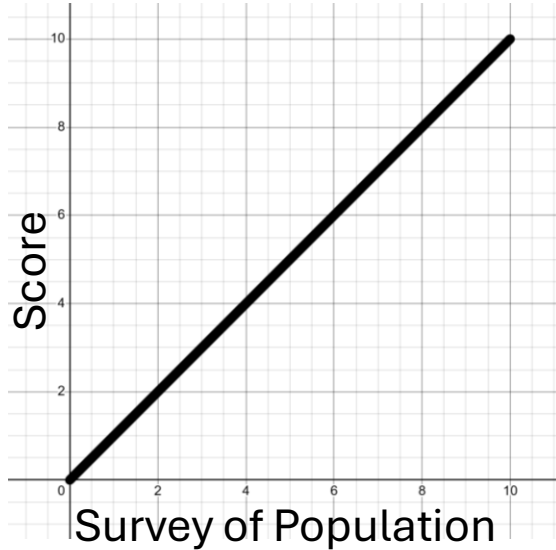
Stakeholders	SH weight	Criteria						
		Perceived Safety	Commute Time	GHG Emissions	Revenue of Businesses	Disruption of construction	Ease of Use for Emergency Services	Cost of Project
Cyclists (Primary)	30.0%	10.0	10.0	9.0	8.0	7.0	7.0	-
City of Vancouver (Primary)	20.0%	9.0	8.0	10.0	9.0	9.0	10.0	10.0
Non-Cycling Car commuter (Primary)	35.0%	9.0	10.0	4.0	-	9.0	4.0	7.0
Translink (Primary)	10.0%	-	9.0	6.0	-	8.0	2.0	-
Business Owners (Secondary)	5.0%	-	6.0	-	10.0	7.0	-	-
Total		7.95	9.30	6.70	4.70	8.20	5.70	4.5
Criteria Weight		16.9%	19.8%	14.3%	10.0%	17.4%	12.1%	9.5%

Appendix B – Weighted Decision Matrix – Evaluation of Solutions (Part 2)

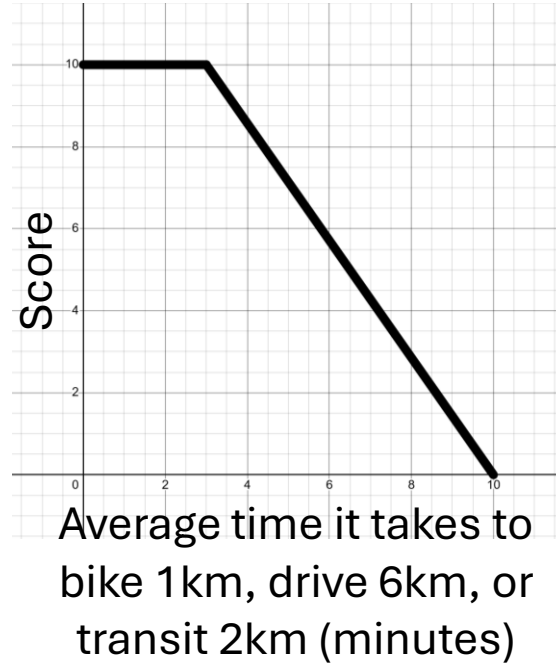
		Kingsway-West (Bollards)		Kingsway-East (Bollards)		Greenway project (Bollards)		Kingsway-West (Concrete Planters)		Kingsway-East (Concrete Planters)		Greenway project (Concrete Planters)	
Criteria	Weighting	Score	Total	Score	Total	Score	Total	Score	Total	Score	Total	Score	Total
Perceived Safety	16.9%	1	0.169	3	0.507	5	0.846	8	1.353	9	1.522	9	1.522
Commute Time	19.8%	7.4	1.464	7.7	1.524	6.4	1.266	7.4	1.464	7.7	1.524	6.4	1.266
GHG Emissions	14.3%	6	0.855	6	0.855	2.5	0.356	6	0.855	6	0.855	2.5	0.356
Revenue of Businesses	10.0%	10	1.000	10	1.000	5	0.500	10	1.000	10	1.000	5	0.500
Disruption of construction	17.4%	2	0.349	2	0.349	5	0.872	2.5	0.436	2.5	0.436	3.5	0.611
Ease of Use for Emergency Services	12.1%	9	1.091	9	1.091	9	1.091	2	0.243	2	0.243	2	0.243
Cost of Project	9.5%	10	0.947	10	0.947	10	0.947	9.7	0.918	9.7	0.918	9.7	0.918
Totals			5.876		6.274		5.879		6.270		6.498		5.417

Appendix C – Approximate Evaluation Criteria Curves (Part 1)

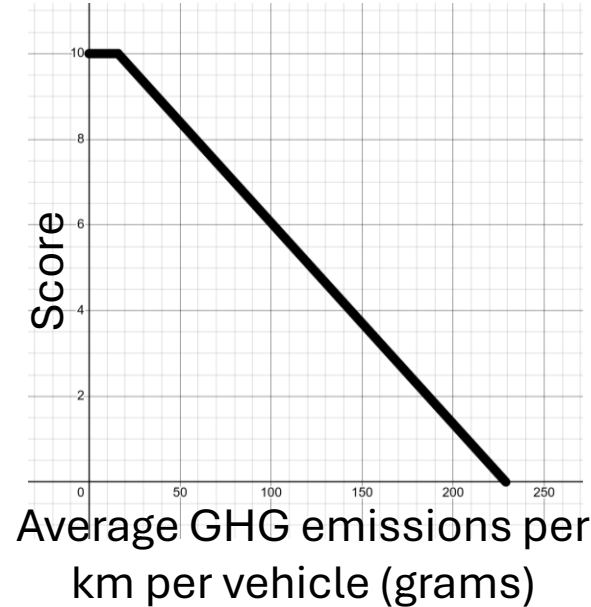
Perceived Safety of Bike Lanes



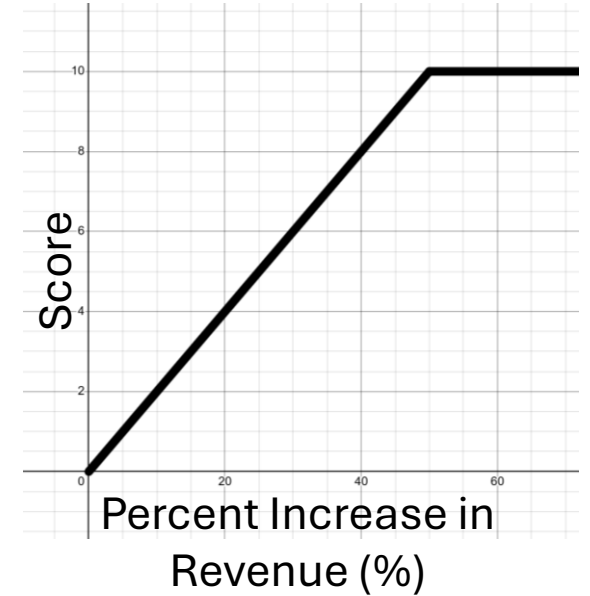
Commute Time



Greenhouse Gas Emissions

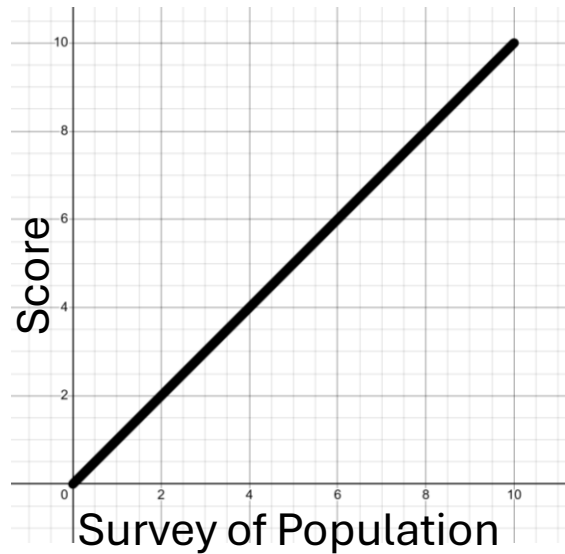


Revenue of Nearby Businesses

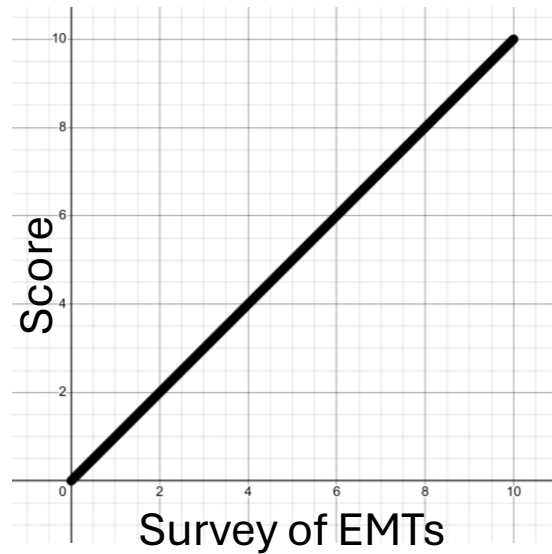


Appendix C – Approximate Evaluation Criteria Curves (Part 2)

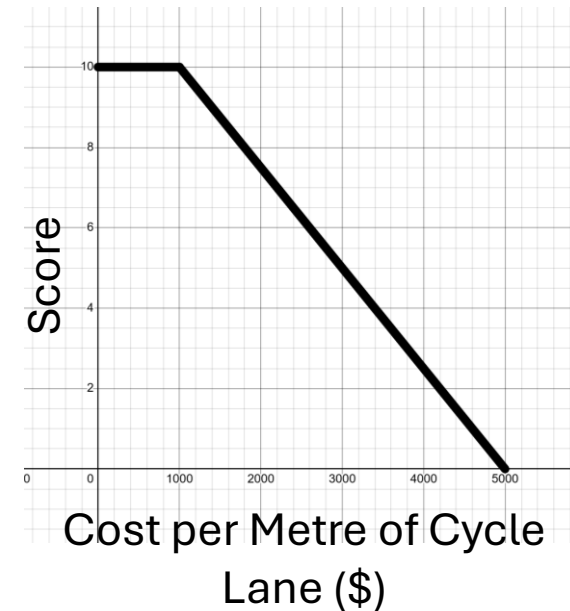
Disruption of Construction



Ease of Use for Emergency Vehicles



Cost of Project



Appendix D - Sources

- Cost of the Project:
 - Cost of cycle Lane with barrier/meter (Benni et al. 2019)
 - Length of each road (Wikipedia, 2025)
- GHG emission:
 - Emission Per Cyclists and Car (European Cyclists' Federation, 2011)
 - Cyclists /Meter Day in Designated Bike Lane (Worst Scenario) (City of Vancouver, 2024)
 - Number of Car on Each Road Per Year (City of Vancouver, 2022)
- Other Criteria:
 - General Consideration.

Appendix E - References

- European Cyclists' Federation. (2011, November). *Cycle more often 2cool down the planet! Quantifying CO₂ savings of cycling* [Report]. Brussels: European Cyclists' Federation.
- Wikipedia contributors. (n.d.). *Kingsway (Vancouver)*. In *Wikipedia*. Retrieved [2025-11-02]
- City of Vancouver. (2024). *How we collect bike volumes on designated bikeways*. Retrieved November 2, 2025
- City of Vancouver. (2022). VanMap interactive map viewer [Web map]. Retrieved November 2, 2025
- Benni, J., Macaraig, M., Malmo-Laycock, J., Smith Lea, N., & Tomalty, R. (2019). *Costing of bicycle infrastructure and programs in Canada* (The Centre for Active Transportation & Clean Air Partnership)